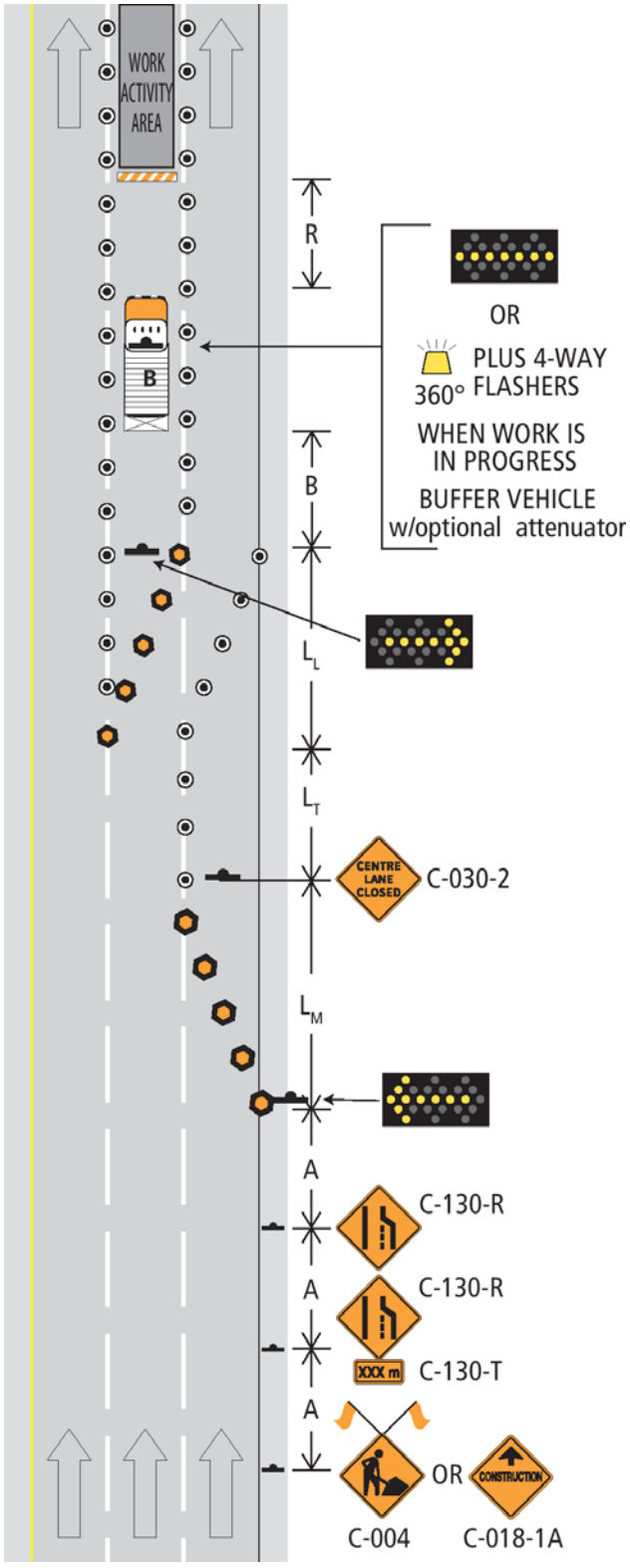
**Figure 8.9: Centre Lane Closure (≥ 70 km/h) – Short and Long Duration**



## 8.10 Centreline Crossover – Short and Long Duration

**Purpose:**

This layout shows the typical setup for the closure of all lanes in an entire direction of travel on a multilane roadway. Traffic from the closed lanes is re-routed across the centreline into the opposing travel lane, thereby reducing the number of lanes available for opposing traffic.

Using this layout reduces road capacity in both directions of travel so it is important to consider the best time of day for implementing this type of closure.

**Standard:**

- A Lane Closed Ahead C-130-L/R sign will be required for the left/right lane closure in advance of the work.
- Traffic shall be merged before it is shifted across the centreline.
- Two-Way Traffic Ahead C-132 signs shall be used in advance of sections where two-way traffic has been established.
- Two-Way Traffic R-010 signs shall be used where two-way traffic has been established.
- Channelizing devices or temporary traffic barriers shall be used to separate opposing lanes of traffic.
- A barricade is required in the last affected lane on the approach side of the work activity area for long-duration work.

**Guidance:**

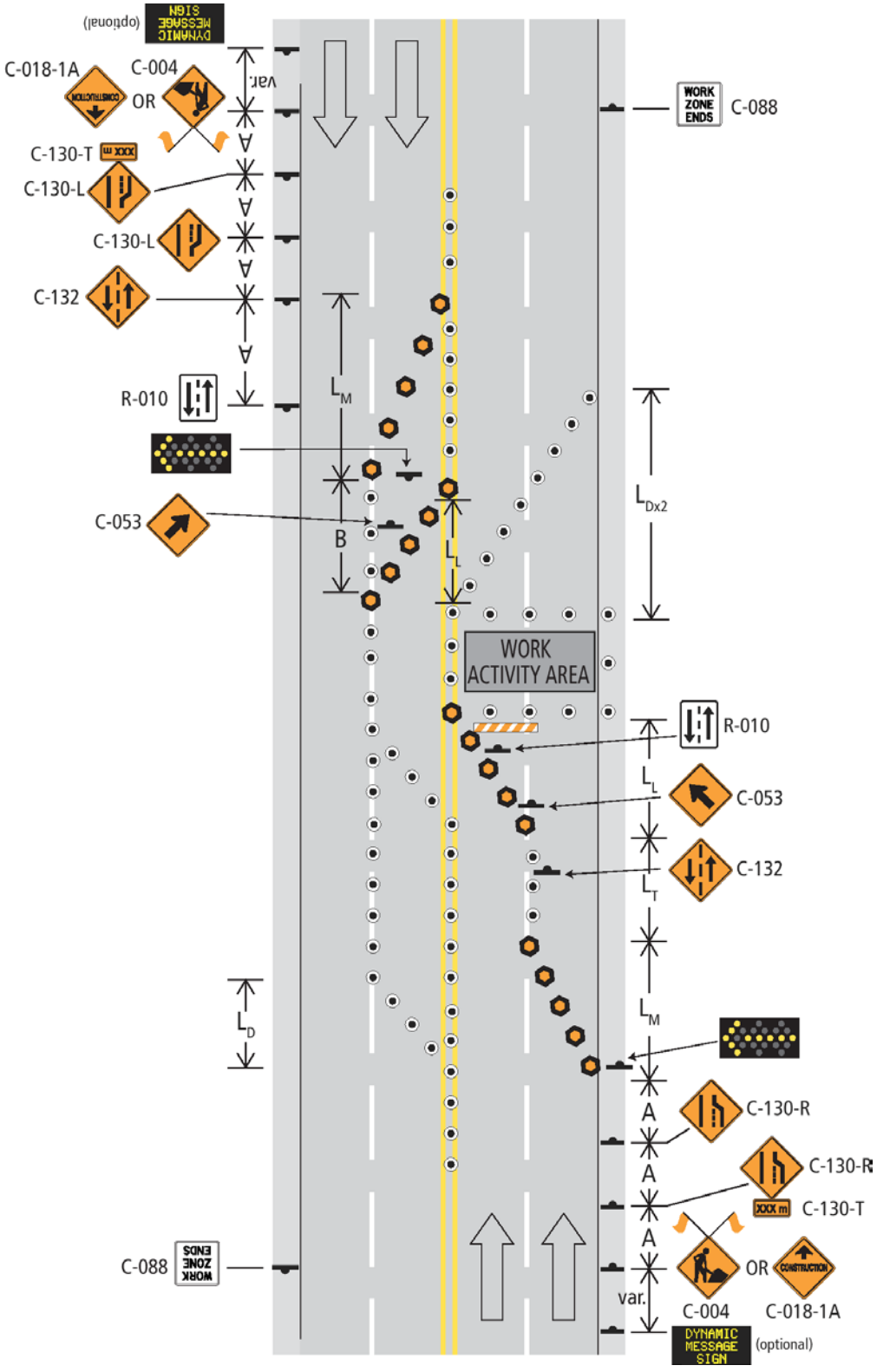
- A dynamic message sign (DMS) is recommended for night work over multiple night-time shifts.
- When re-establishing normal traffic flow, remove the signs and devices for the diverted lanes first, and then remove the devices for the opposing lanes.

**Options:**

- If the speed limit is ≤ 60 km/h, the upstream Right Lane Closed Ahead C-130-R sign with distance tab may be omitted, and the Construction Ahead C-018-1A sign moved downstream by Table B Distance A.
- A custom Traffic Pattern Changed Zx-030 sign with an implementation plan on counterflow operations and transitions may be used as a supplement. See Zx-030 *Appendix B*.
- The flashing arrow board (FAB) may be replaced as shown below.

| Speed Limit | Workers or Work Vehicles Present | No Workers or Vehicles Present |
|---|---|---|
| ≤ 60 km/h | Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers. | Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light. |
| ≥ 70 km/h | No substitution. | |

**Figure 8.10: Centreline Crossover – Short and Long Duration**



## 8.11 Two-Way Left-Turn Lane Closed – Short and Long Duration

**Purpose:**

This layout shows the typical setup for the closure of a two-way left-turn lane. If the work will encroach into either travel lane, a left lane closure for the affected lane(s) should be considered.

**Standard:**

- A flashing arrow board (FAB) in caution mode is required for each direction of travel.
- Barricades are required at each end of the work activity area for long-duration work.

**Guidance:**

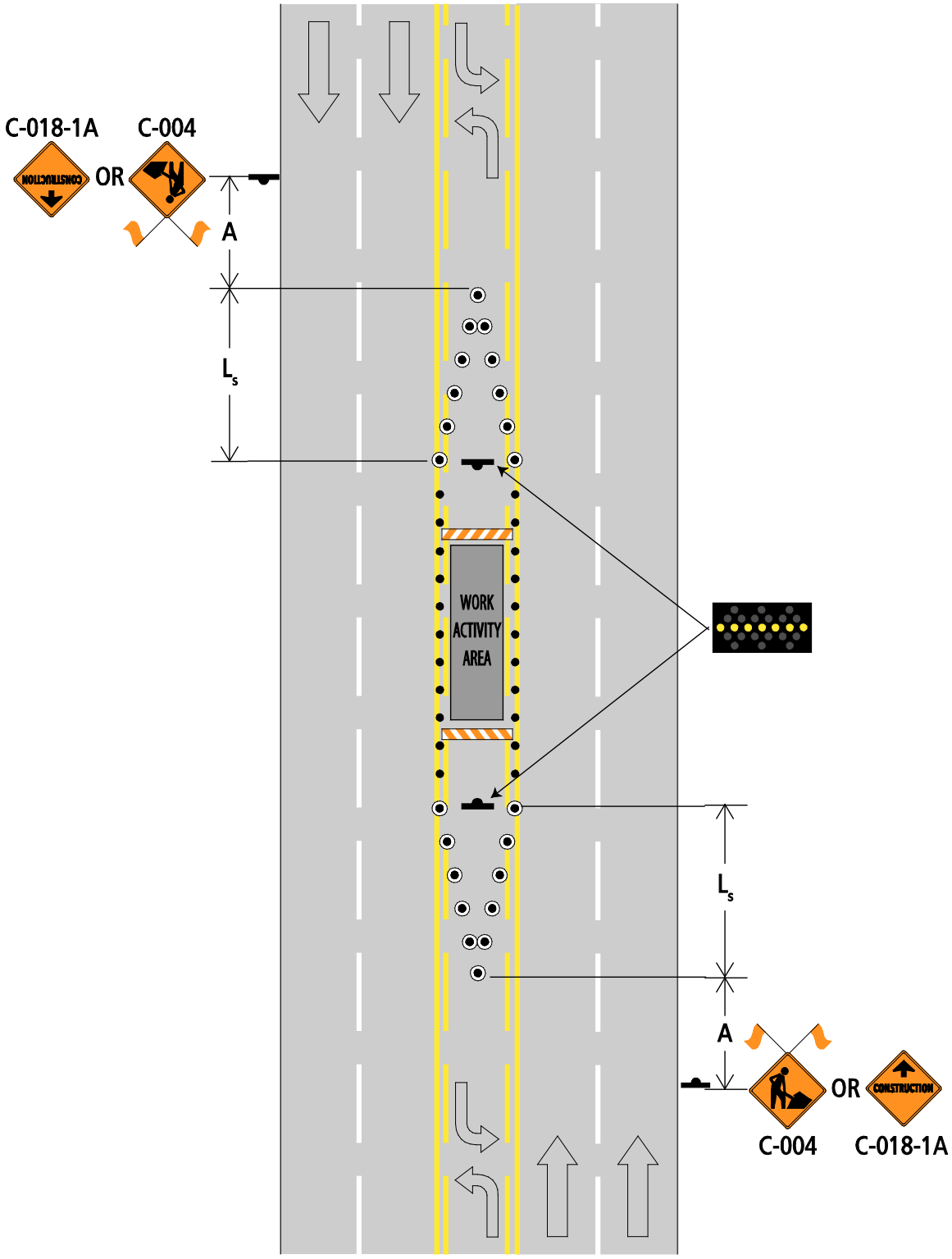
- Left-turning movements may be prohibited along the work activity area and taper.

**Options:**

- Depending on the nature of the work and whether or not it will encroach into either travel lane, one or both adjacent lanes may also have to be closed.
- For speeds ≤ 60 km/h, cones may be used instead of the tubular markers to channelize traffic.
- The flashing arrow board (FAB) may be replaced as shown below.

| Speed Limit | Workers or Work Vehicles Present | No Workers or Vehicles Present |
|---|---|---|
| ≤ 60 km/h | Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers. | Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light. |
| ≥ 70 km/h | No substitution. | |

**Figure 8.11: Two-Way Left-Turn Lane Closed – Short and Long Duration**



## 8.12 Runaway Lane Open – Short and Long Duration

**Purpose:**

Runaway lanes are emergency escape ramps created for long, descending grades. On such grades, vehicle speeds may increase and brakes may overheat and fail because of the extensive braking used to slow the vehicle. The runaway lanes allow vehicles with brake problems to exit the roadway and stop. Runaway lanes should be kept open whenever possible because of their importance for roadway safety.

This layout shows the typical setup for work adjacent to a runaway lane that remains open. Effort should be made to **clearly communicate the open status of the runaway lane** to truck drivers and other approaching motorists.

**Standard:**

- Construction equipment shall never be left parked in—or blocking access to—an open runaway lane.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the exit ramp.
- A flashing arrow board (FAB) shall be placed inside the taper.
- A buffer space shall always be used.
- A barricade is required on the approach side of the work activity area for long-duration work.
- A dynamic message sign (DMS) shall be used to communicate the runaway lane status.

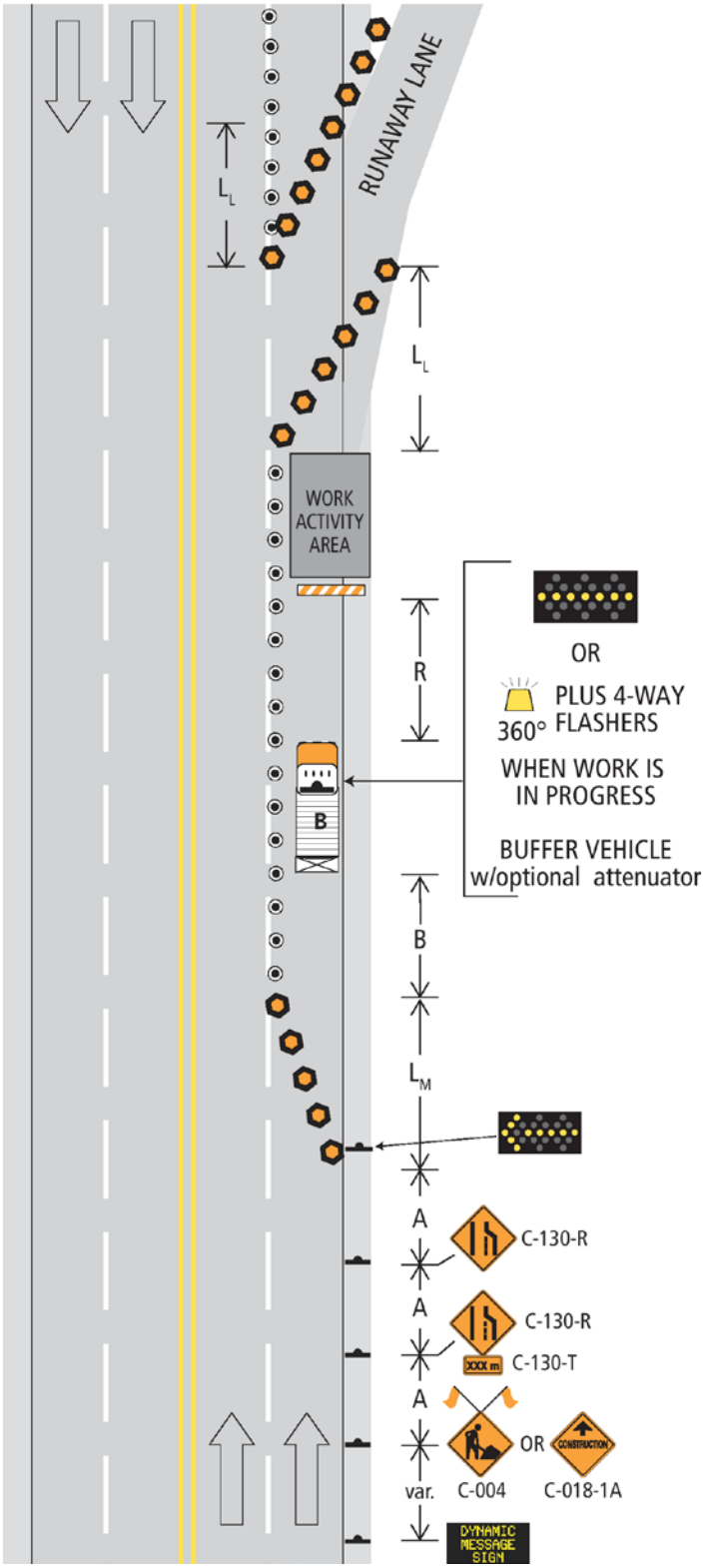
**Guidance:**

- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.
- Vehicles should not be queued in advance of—or through—the entrance to a runaway lane.

**Options:**

- None at this time.

**Figure 8.12: Runaway Lane Open – Short and Long Duration**



## 8.13 Runaway Lane Closed – Short and Long Duration

**Purpose:**

Runaway lanes are emergency escape ramps created for long, descending grades. On such grades, vehicle speeds may increase and brakes may overheat and fail because of the extensive braking used to slow the vehicle. The runaway lanes allow vehicles with brake problems to exit the roadway and stop. Runaway lanes should be kept open whenever possible because of their importance for roadway safety.

This layout shows the typical setup for work in front of a runaway lane that is closed. When work is occurring in front of runaway lanes, effort shall be made to clearly communicate the closed status of the runaway lane to truck drivers and other approaching motorists.

Even if a runaway lane is closed, extreme caution should be used when working in front of or across the runaway lane. Never assume that just because it is considered closed, a truck would not attempt to use it in an emergency.

**Standard:**

- Truck drivers shall be notified of runaway lane closures at (within or in advance of) the brake check and in advance of the grade.
- A Right Lane Closed Ahead C-130-R sign with a C-130-T distance tab and a second Right Lane Closed Ahead C-130-R sign shall be placed in advance of the entrance ramp.
- The Runaway Lane Closed C-067 sign shall be positioned 200 metres or distance A, whichever is greater, in advance of the affected runaway lane. It shall be removed or covered as soon as possible once the runaway lane is available.
- For construction zones across runaway lanes, use the Closed C-061 tab mounted above or across the existing runaway signs.
- A dynamic message sign (DMS) shall be used to communicate the runaway lane closure. Appropriate positioning and messaging should be discussed with the Road Authority.
- A buffer space shall always be used.
- A barricade is required on the approach side of the work activity area for long-duration work.

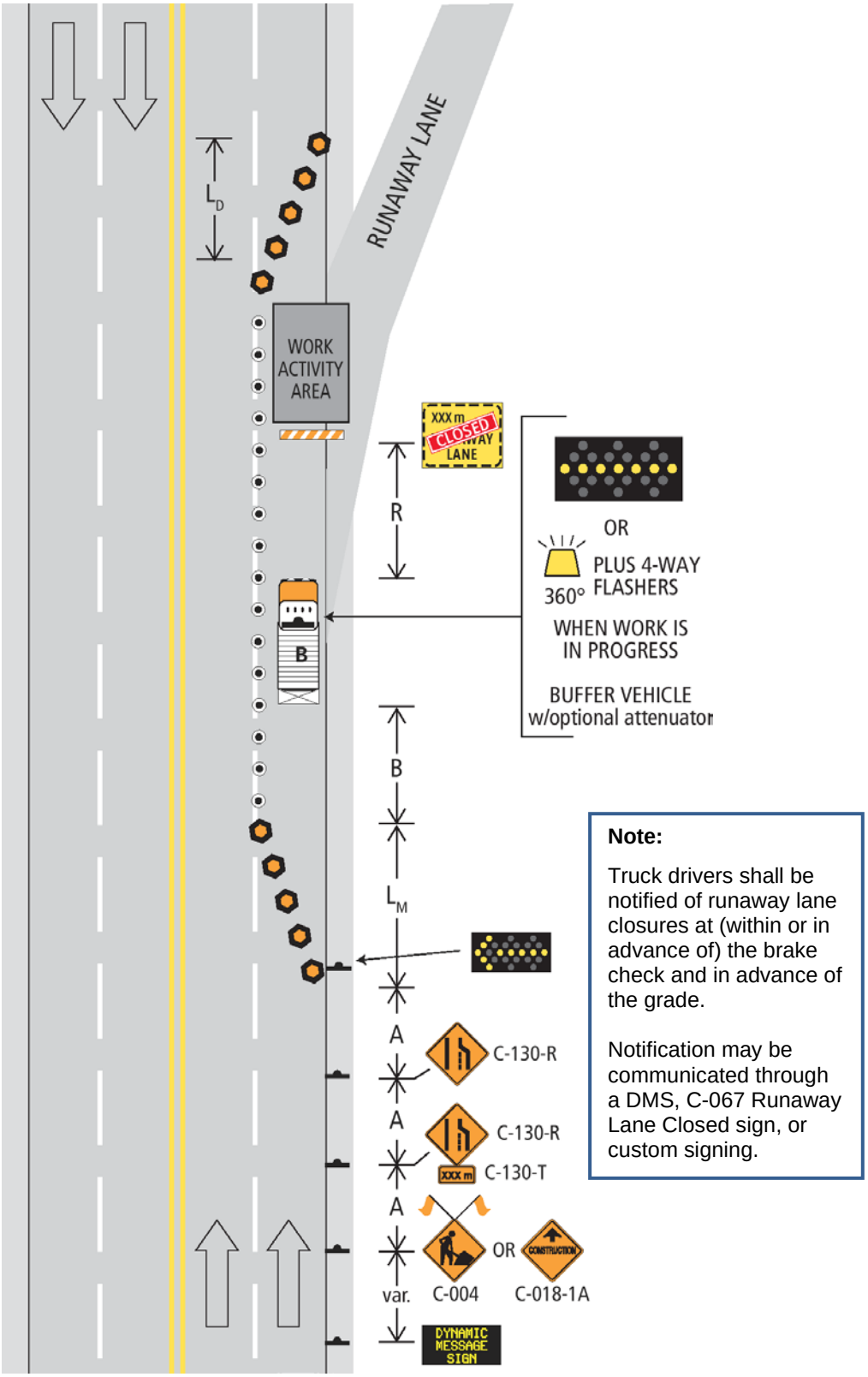
**Guidance:**

- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.
- Vehicles should not be queued in advance of—or through—the entrance to a runaway lane.
- If a lane closure is not required, the Right Lane Closed Ahead C-130 series and flashing arrow board (FAB) can be removed.

**Options:**

- Notification at the brake check may be communicated through a dynamic message sign (DMS), C-067 Runaway Lane Closed sign, or custom signing.

**Figure 8.13: Runaway Lane Closed – Short and Long Duration**




**Note:**

Truck drivers shall be notified of runaway lane closures at (within or in advance of) the brake check and in advance of the grade.

Notification may be communicated through a DMS, C-067 Runaway Lane Closed sign, or custom signing.

## 8.14 Passing/Climbing Lanes – Lane Shift – Short and Long Duration

**Purpose:**

In mountainous areas, the presence of truck climbing lanes or uphill passing lanes provides an opportunity to avoid single lane alternating traffic control when a lane closure is necessary in the downhill lane.

A lane shift may be used to move downhill traffic into the left-most uphill lane while uphill traffic is restricted to a single lane. Using a lane shift is preferable to stopping downhill traffic, but it may not always be feasible.

**Standard:**

- Two-Way Traffic Ahead C-132 signs shall be used in advance of sections where two-way traffic has been established.
- Two-Way Traffic R-010 signs shall be used along tangents where two-way traffic has been established.
- A flashing arrow board (FAB) shall be placed inside the tapers.
- Passing/climbing lane signs that are no longer applicable because of the lane shift shall be covered.
- A barricade is required on the approach side of the work activity area for long-duration work.

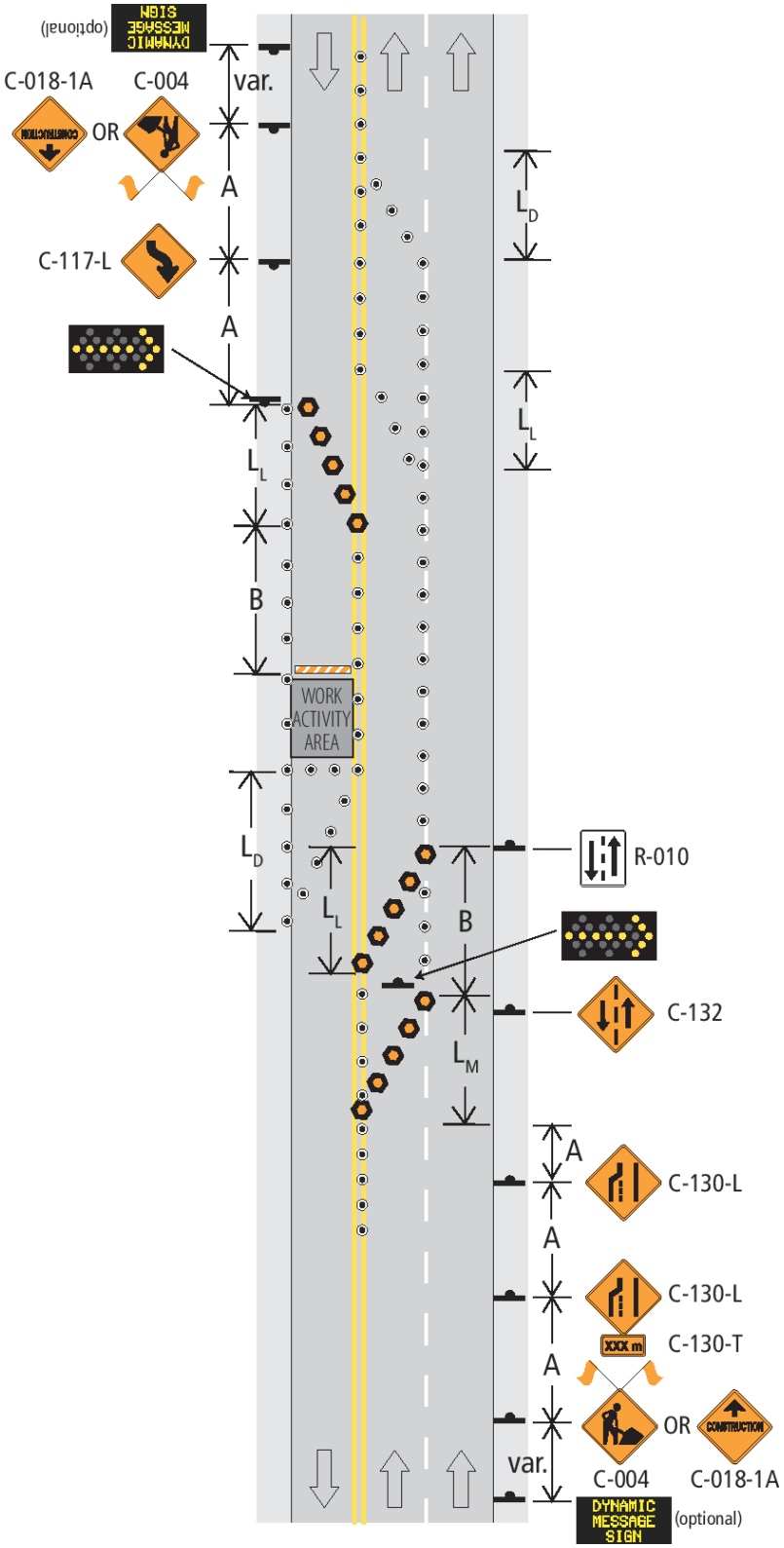
**Guidance:**

- Passing restrictions in both directions should be discussed with the Road Authority to determine whether or not traffic volumes will allow decreased capacity in the uphill direction.

**Options:**

- None at this time.

**Figure 8.14: Passing/Climbing Lanes – Lane Shift – Short and Long Duration**



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# PART B – TRAFFIC CONTROL

## Section 9: Traffic Control Layouts – Multilane Divided Roadways

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# PART B – TRAFFIC CONTROL

## Section 9: Traffic Control Layouts – Multilane Divided Roadways

| LEGEND | | | |
|---|---|---|---|
| | Flexible Drum | | Paint Truck |
| | Tubular Marker | | Escort Truck |
| | Cone | | Chaser Vehicle |
| | Sign | | Vehicle Mounted Rear Crash Attenuator |
| | Traffic Control Person | | |
| | Work Activity Area | | 360° Flashing Light |
| | Work Truck | | Portable Traffic Signal |
| | Shadow Vehicle | | Barricade and Fencing |
| | Shadow Vehicle #1 | | Flashing Arrow Board (FAB) |
| | Shadow Vehicle #2 | | Flashing Arrow Board (FAB) in caution mode |
| | Buffer Vehicle | | Dynamic Message Sign (DMS) |

**TABLE A – TAPER LENGTHS**

| Taper Types (m) | | Regular Posted Speed Limit (km/h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | ≤50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |
| Merge Taper Length | $L_M$ | 35 | 55 | 160 | 190 | 210 | 230 | 250 | 280 |
| Lane Shift Taper Length | $L_L$ | 30 | 50 | 80 | 100 | 110 | 120 | 130 | 140 |
| Downstream Taper Length | $L_D$ | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 |
| TCP, Signal, and Shoulder Taper Length (min. 5 devices) | $L_S$ | 5 | 8 | 15 | 15 | 15 | 15 | 15 | 15 |
| Minimum Tangent Length between Tapers | $L_T$ | 30 | 60 | 160 | 190 | 210 | 230 | 250 | 280 |
| Run-In Length on Centreline | $L_R$ | 40 | 50 | 60 | 60 | 70 | 80 | 90 | 100 |

**TABLE B – DEVICE SPACING LENGTHS**

| Device Spacing (m) | | Regular Posted Speed Limit (km/h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | ≤50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |
| Construction Sign Spacing | A | 40 | 60 | 80 | 100 | 150 | 150 | 200 | 200 |
| Buffer Space | B | 30 | 40 | 60 | 80 | 110 | 140 | 170 | 200 |
| Roll-Ahead Buffer Distance | R | 30 | 30 | 40 | 40 | 40 | 50 | 50 | 50 |
| Channelizing Device Spacing for Tapers | C | 10 | 10 | 15 | 15 | 15 | 15 | 15 | 15 |
| Channelizing Device Spacing on Curves and Tangents | D | 10 | 10 | 30 | 30 | 40 | 40 | 40 | 50 |

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## 9.1 General Information – Multilane Divided Roadway

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

**Standard:**

- For short-duration work, a Crew Working Ahead C-004 sign shall be used.
- For long-duration work, a Construction Ahead C-018-1A sign shall be used.
- Where the speed limit is ≥ 70 km/h, a buffer space shall be used.
- A flashing arrow board (FAB) shall be used when a lane is closed.
- All work, buffer, and shadow vehicles shall be equipped with a 360-degree flashing light and 4-way flashers.
- A barricade is required in the last affected lane on the approach side of the work activity area for long-duration work.

**Guidance:**

- Where the speed limit is ≥ 70 km/h a buffer vehicle should be used when workers are present.
- A vehicle-mounted crash attenuator is recommended for use on buffer vehicles.

**Options:**

- Where cyclists are regularly observed using the shoulder, a Share the Road W-132-1 sign may be used with an appropriate cycling hazard tab (see *Section 18: Traffic Control Layouts – Bicycle Lanes* for additional information).
- A portable dynamic message sign (DMS) may be used to provide advance messaging for drivers.

## 9.2 Typical Construction Speed Zone Signing – Multilane Divided Roadway

**Purpose:**

Construction speed limits are regulatory speeds established in Construction Speed Zones within long-term construction and maintenance project areas where there are continuous hazards for motorists or where workers are in close proximity to active travel lanes.

Construction Speed Zones should be applied prudently because **overuse reduces effectiveness.** Drivers should be able to perceive the need to reduce speed.

**Standard:**

- Conflicting speed limit signs within the Construction Speed Zone shall be covered or removed.

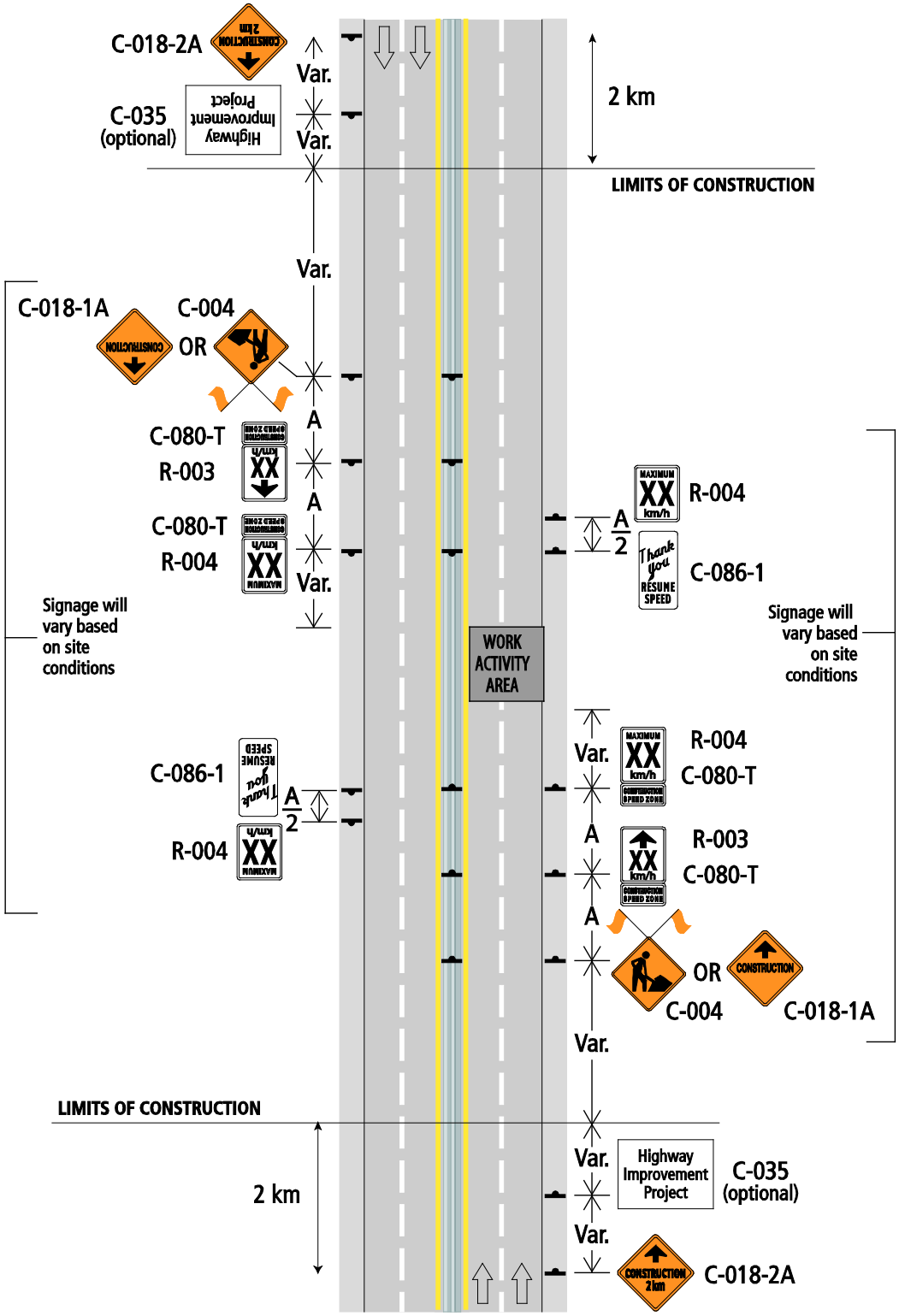
**Guidance:**

- Signage should be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- The Maximum Speed R-004 speed sign with a Construction Speed Zone C-080-T distance tab should be positioned at the beginning of the active work area.
- Construction speed limit signs should be covered or removed when no work is occurring and other hazards are not present.
- It is important to record when construction speed limits are installed and covered/removed.
- See also *Section 2.4: Management of Speed* for information on using Construction Speed Zones.

**Options:**

- Use of the Construction Project C-035 sign is project-dependent. It is typically used only for large projects. See *Appendix B: Standard Construction Signs* for information on using this sign.
- If secondary signs are applied, they should be installed on the left side of the highway.

**Figure 9.2: Typical Construction Speed Zone Signing – Multilane Divided Roadway**



## 9.3 Emergent Work (<5 Minutes) – Multilane Divided Roadway

**Purpose:**

Emergent work involves very short-duration activities for which setting up and taking down temporary traffic control devices may take more time than the actual work, and expose workers to greater risk. Each entry onto the travelled portion of the roadway lasts less than 1 minute, and the total time to complete the task is less than 5 minutes.

This does not include emergencies, which are situations which require immediate response to save lives or prevent serious injury using whatever resources are available.

It may occur when an unanticipated situation or event—one that presents a risk to the travelling public—is discovered during travel or work activities. It can be considered unplanned, urgent maintenance work. A Traffic Control Plan is not required for emergent work.

Emergent work may include the removal of debris from the roadway (e.g., tree limbs, lost cargo, dead animals, tire and other vehicle debris, and the manual removal of rock). Isolated pothole patching—patching 1 or 2 potholes in a 1-kilometre section of road—may be considered emergent or brief-duration work. However, it is mobile work when the work crew has to stop several times in succession within a 1-kilometre section to patch multiple potholes.

**Standard:**

- A risk evaluation is required to determine whether or not the work activity is considered emergent (see *Table C – Risk Evaluation for Emergent or Brief-Duration Work* in *Section 6.10* or *Appendix F*).
- The entire work activity shall be completed in less than 5 minutes.
- The work vehicle shall use a flashing arrow board (FAB) in caution mode or a 360-degree flashing light and 4-way flashers.

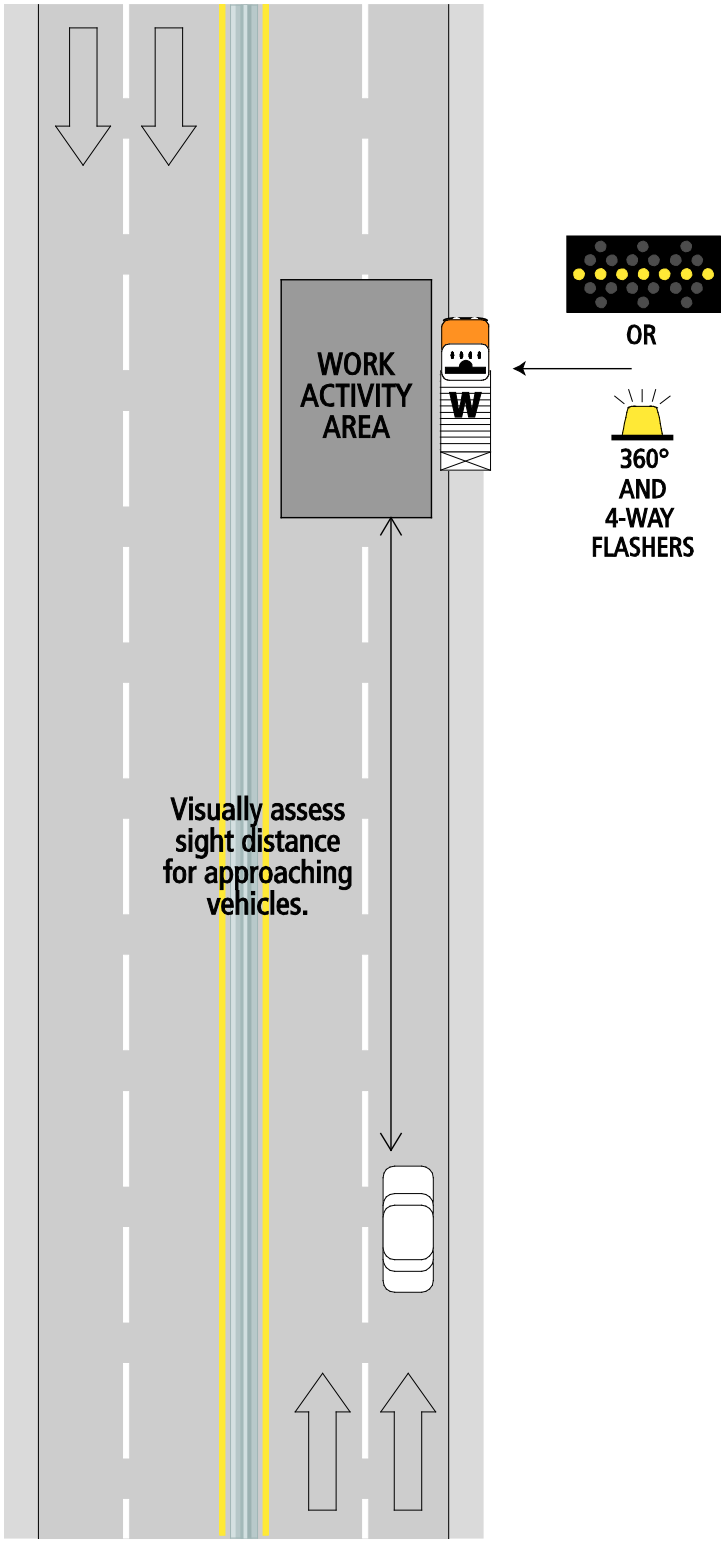
**Guidance:**

- If a dynamic message sign (DMS) or flashing arrow board (FAB) is used for an operation on the shoulder, it should be set to caution mode.
- Alternatively, a 360-degree flashing light and 4-way flashers should be used.
- The position of the work vehicle in relation to the work area may be adjusted based on the available sight distance, shoulder/off-roadway conditions, and work being performed.

**Options:**

- None at this time

**Figure 9.3: Emergent Work (<5 Minutes) – Multilane Divided Roadway**



## 9.4 Brief-Duration Work (<15 Minutes) – Multilane Divided Roadway

**Purpose:**

Brief-duration work is generally planned, although the exact location or extent of the work required may not be fully known. It requires less than 15 minutes to complete.

Brief-duration work may include:

- locating drainage structures or other roadway features or components
- cleanup of material spills and removing debris from the roadway (e.g., small fallen trees, larger tree limbs, crash debris, etc.)
- quick repairs intended as a partial or temporary response to damage or failure

If the work is expected to last 15 minutes or longer, additional resources will be needed to implement a short-duration work zone. The work should be delayed until the appropriate work zone equipment and devices are available.

Isolated pothole patching—patching 1 or 2 potholes in a 1-kilometre section of road—may be considered emergent or brief-duration work. However, it is mobile work when the work crew has to stop several times in succession within a 1-kilometre section to patch multiple potholes.

**Standard:**

- A Crew Working Ahead C-004 is required in advance of the work.
- A risk evaluation is required to determine whether or not the work activity is considered emergent (see *Table C – Risk Evaluation for Emergent or Brief-Duration Work* in *Section 6.10* or *Appendix F).*
- The work activity shall be completed in less than 15 minutes.
- The work vehicle shall use a flashing arrow board (FAB) in caution mode or a 360-degree flashing light and 4-way flashers.

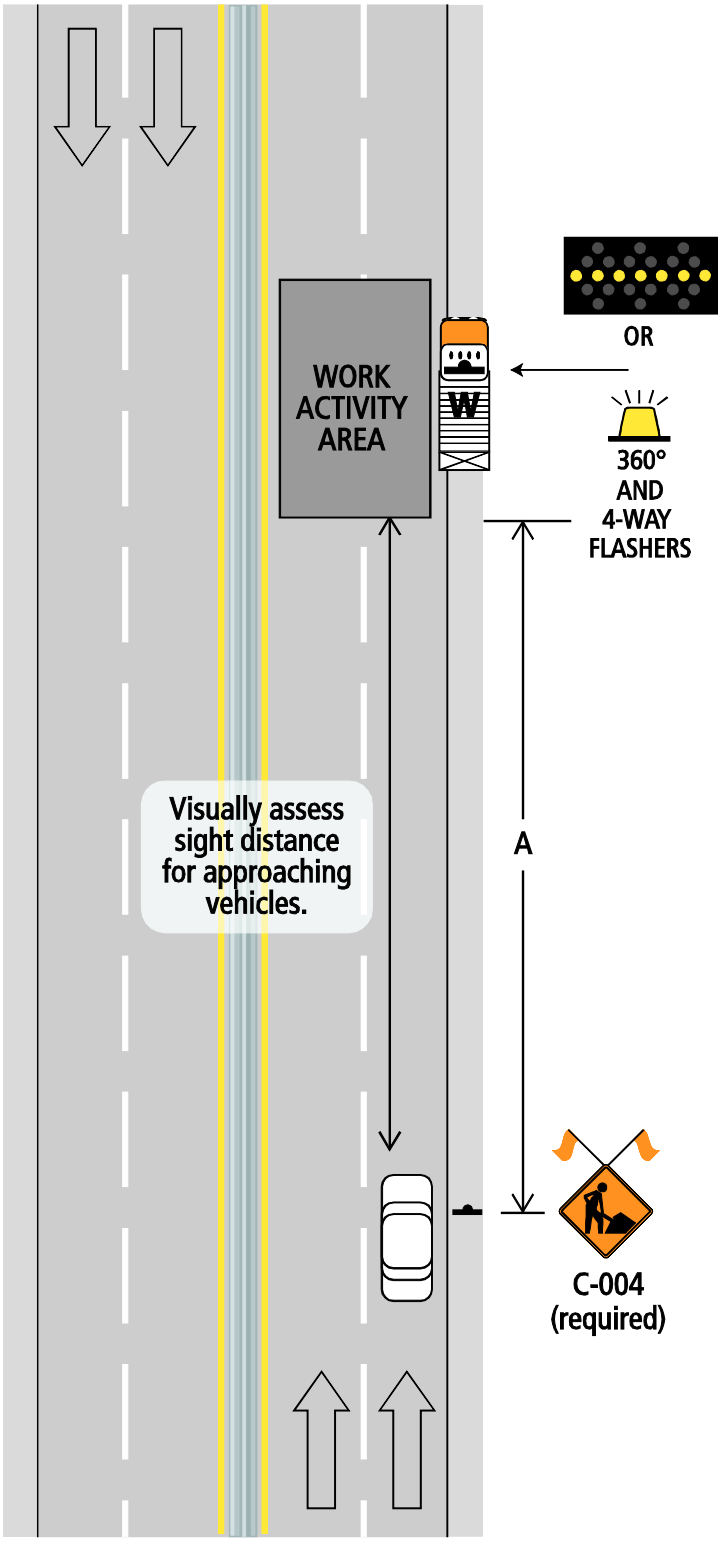
**Guidance:**

- If a dynamic message sign (DMS) or flashing arrow board (FAB) is used for an operation on the shoulder, it should be set to caution mode.
- The position of the work vehicle in relation to the work area may be adjusted based on the available sight distance, shoulder/off-roadway conditions, and the work being performed.

**Options:**

- None at this time.

**Figure 9.4: Brief-Duration Work (<15 Minutes) – Multilane Divided Roadway**



## 9.5 Work on Shoulder – Short and Long Duration

**Purpose:**

Stationary work on the shoulder takes place outside the travel lanes but occupies part or all of the shoulder area.

If shoulder work encroaches into a travel lane, a full lane closure shall be implemented.

**Standard:**

- When work is in progress and workers are present, a flashing arrow board (FAB) in caution mode is required in the taper or immediately in advance of the work area.
- A barricade is required on the approach side of the work activity area for long-duration work.

**Guidance:**

- When work is not in progress but the work area has not been cleared, care should be taken to isolate it from the travelled roadway.

**Options:**

- Signage may be installed along the divided median. If the median space is limited, smaller sized signs may be used.
- Advance warning signs may be applied in the opposing direction of travel.
- If the speed limit is ≤ 60 km/h, or there is insufficient room, tubular markers may replace drums for the taper.
- The channelizing devices used alongside the work area may be tubular markers or cones.
- Advance warning signs may be omitted for short-duration work if the shadow vehicle displays a vehicle-mounted dynamic message sign (DMS) or flashing arrow board (FAB) in caution mode and uses a 360-degree flashing light and 4-way flashers.

**Figure 9.5: Work on Shoulder – Short and Long Duration**

